

Cleaning Irish Used-Car Listings

Using association rule mining to decide which attributes were worth cleaning

Data Warehousing & Mining | Topic 4: Data Cleaning

Association rule mining

The problem

- Web scraping is slow, and every attribute we clean is time costly
- Ads have too attributes: specs, condition, NCT, warranty, location etc.
- Cleaning all of them would waste time on ones that don’t affect price

Why it fit

- Finds rules of the form “if cars differ in X, then price differs”
- Works on categorical attributes like trim, county and seller type
- Gives every attribute one comparable score, so we can rank them and remove attributes that have minimal affect on price

Calculating Lift Value



Car A

Same make, model, year, trim, mileage band…

Seller: dealer

VS



Car B

Same make, model, year, trim, mileage band…

Seller: private

Only one attribute differs, so any price gap can be put down to that attribute, not to age or mileage.

Lift = confidence ÷ base

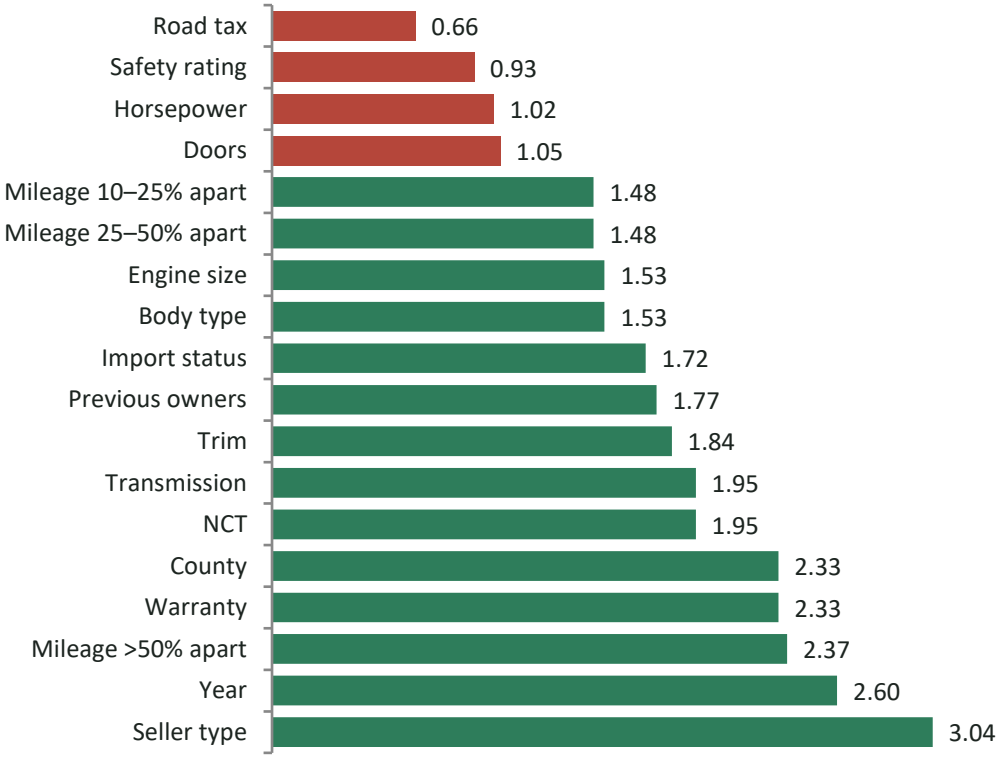
Base: how often the price outcome happens across all cars

Confidence: how often it happens across pairs that differ in the attribute

Lift above 1: the attribute makes the outcome more likely; the higher, the stronger

Seller type: 58% confidence ÷ ~19% base = lift 3.04

Rule Mining: Outcomes



Cut before cleaning

The red bars: lift near or below 1, so a difference in these barely changed the price. We never cleaned them.

3.04 **Seller type, the top attribute**

In 80% of pairs dealer cars were more expensive.
Year had the next highest lift value at 2.60.

Used Cleaning Methods

	Method	When	Why
1	**Ignore the tuple**	No price or “POA”; mileage missing and not in the description	*Price is the outcome, so these rows can’t be used*
2	**Fill in manually**	Missing engine size found in title (3), transmission found in text or photos (6), Odd prices and mileages included real values in the description	*Few enough cases to check by hand*
3	**Global constant**	Trim = Unknown, import = Unknown, NCT = Unknown (older cars), N/A (new cars)	*No reliable way to guess; keep the row*
4	**Median by year**	Previous owners filled with the median for that year	*Older cars generally have more owners*
5	**Most probable value**	Assume seller type is private, unless the ad offers a warranty	*A warranty generally means a dealer*

Dealing with Outliers



Fake values

- Price of €1, €1234, €123456 etc. used as a placeholder
- Mileage of 1234 km etc. typed to fill a required field
- No real car has these. We recovered the true value from the ad description when possible



N/A vs Unknown

- NCT = N/A for new cars: not yet due, so nothing is missing
- NCT = Unknown for older cars: due, but not listed
- Two separate values keep that difference in the data

Outcomes

4

attributes cut before any cleaning, saving scraping and cleaning time

1.5%

Removed 1.5% of tuples from our dataset

851

Changed the value of 851 attributes out of 25,420